



Multi-Tenant Campus Club & Event Management SaaS

Complete File-by-File Technical Guide & Interview Master Cheat-Sheet

(Har file ka Role, Kab & Kyon use hoti hai, aur Interview me kaise explain karein)



1. Project Overview & Quick Elevator Pitch

Project Ka Purpose: Yeh ek Multi-Tenant SaaS platform hai jisme har College Club (jaise Dance Club, Coding Club) ko apna alag branded sub-portal milta hai. Students events me register karke cryptographic QR Gate Pass paate hain, volunteers mobile camera se gate par instant scan karke duplicate entry rokhte hain, aur post-event 1-click me high-resolution standard A4 completion certificates auto-dispatch hote hain.



2. Backend Files Breakdown (Server, Database & APIs)



`backend/server.js`

BACKEND CORE ENGINE

- 📌 **Kya hai:** Express application ka main entry point jo backend server ko initialize aur run karta hai.
- 🕒 **Kab chalta hai:** Jab command `node server.js` execute hota hai (Port 5000 par start hota hai).
- ⚙️ **Kaise kaam karta hai:** Yeh CORS enable karta hai, `express.json({ limit: '25mb' })` body size limit set karta hai (taaki certificates ke high-res images pass ho sakein), MongoDB database connect karta hai, aur sabhi API routes ko mount karta hai.
- ❓ **Kyon zaroori hai:** Bina `server.js` ke frontend ko API endpoints aur database connectivity nahi mil sakti.



Interview Tip: Bolen: "*server.js acts as our centralized API Gateway, mounting all multi-tenant modular route controllers and configuring 25MB body payload support for custom certificate image templates.*"

backend/models/Tenant.js

MONGOOSE SCHEMA

✦ **Kya hai:** College Clubs (Tenants) ka Database Schema / Data Structure definition.

⚙️ **Key Fields:** name (Club ka naam), slug (Unique URL ID jaise 'dance-club'), primaryColor & secondaryColor (Branding theme), certificateTemplateUrl (Custom A4 Frame image), signatory1Name/Title & signatory2Name/Title (Signatures on certificates), status ('ACTIVE' ya 'SUSPENDED').

❓ **Kyon zaroori hai:** Yeh Multi-Tenancy ka core model hai jo har club ki identity aur certificate styling store karta hai.

 **Interview Tip:** Bolen: "The Tenant model implements dynamic tenant customization, storing brand colors, unique route slugs, and custom certificate template metadata."

backend/models/Event.js

MONGOOSE SCHEMA

✦ **Kya hai:** Clubs dwara host kiye jaane wale Events ka Data Schema.

⚙️ **Key Fields:** tenantId (Kis club ka event hai), title, description, date, location, capacity (Total seats), registrationCount (Kitne passes book hue), ticketPrice (₹ 0 for Free ya Paid).

❓ **Kyon zaroori hai:** Real-time seat tracking aur paid/free ticketing manage karne ke liye.

backend/models/Registration.js

MONGOOSE SCHEMA

✦ **Kya hai:** Student Entry Passes aur Issued Certificates ka record.

⚙️ **Key Fields:** tenantId, eventId, name, email, department, passId (Unique passcode), qrCode (Base64 QR image), checkedIn (Boolean - Gate par scan hua ya nahi), checkedInAt (Timestamp), certificateIssued, certificateHash (Anti-fraud cryptographic hash).

❓ **Kyon zaroori hai:** Duplicate entry rokne aur certificate authenticity verify karne ka main database table hai.

 **Interview Tip:** Bolen: "The Registration schema maintains the atomic check-in state (`checkedIn`, `checkedInAt`) and stores cryptographic verification hashes to prevent fraud."

backend/middleware/auth.js & tenant.js

SECURITY & ROUTING

✦ **Kya hai:** Security guards jo verify karte hain ki request authenticated hai ya nahi, aur request kis club ki hai.

⚙️ **Kaise kaam karta hai:** auth.js HTTP request headers me se JWT Token check karta hai. tenant.js URL slug (e.g. /club/dance-club) se club dhoondh kar req.tenant me inject karta hai.

❓ **Kyon zaroori hai:** Taaki unauthorized log admin dashboard access na kar sakein aur ek club ka data doosre club me mix na ho.



📌 **Kya hai:** Sabse powerful backend controller jisme Ticketing, Gate Scanner Validation, Custom Certificate Settings aur Vector SVG Generation Logic hai.

⚙️ **Key APIs:**

- POST /api/club/:slug/registrations - Student pass booking & QR generation.
- POST /api/club/:slug/scanner/validate - Gate camera verification & duplicate check.
- GET/PUT /api/club/:slug/certificate-settings - Save custom background frame & signatures.
- POST /api/club/:slug/registrations/:id/send-certificate - High-res standard A4 (1414 x 1000) vector SVG render karke direct email attachment dispatch karna.



Interview Tip: Bolen: *"This controller implements our SVG Vector Rendering Engine, calculating mathematical coordinate placements for dynamic student names, gold authenticity seals, and custom background templates."*



3. Frontend Files Breakdown (React 18 + Vite SPA)



`frontend/src/App.jsx`

CLIENT ROUTING



Kya hai: Frontend application ka central routing hub (React Router v6).



Route Mapping:

- `/` → **Home.jsx** (Campus Platform Hub)
- `/club/:slug` → **ClubHome.jsx** (Public Club Portal)
- `/club/:slug/admin/dashboard` → **ClubAdminDashboard.jsx** (Organizer Console)
- `/club/:slug/admin/scanner` → **ClubAdminScanner.jsx** (Gate Camera Scanner)
- `/certificate/:id` → **CertificateView.jsx** (E-Certificate View & Print)
- `/super-admin/dashboard` → **SuperAdminDashboard.jsx** (Campus Master Console)



Kyon zaroori hai: Single Page App me fast page transitions aur dynamic multi-tenant routes provide karta hai.



`frontend/src/pages/Home.jsx`

PLATFORM HUB



Kya hai: Main Landing Page jahan sabhi clubs list hote hain aur naya club launch hota hai.



Key Features: Live Search & Filter, "+ New Club" creation modal, aur **Direct Club Link QR Code Generator** (Google Lens se direct scan karne ke liye).



`frontend/src/pages/ClubHome.jsx`

TENANT PORTAL



Kya hai: Kisi specific club ka public homepage (e.g. `/club/dance-club`).



Key Features: Club branding theme, live active events feed, seat capacity progress bar, instant QR pass reservation button, aur Club Share QR modal.



`frontend/src/pages/ClubAdminDashboard.jsx`

ORGANIZER CONSOLE



Kya hai: Club Organizer ka main management dashboard.



4 Main Tabs:

-  **Analytics & Overview:** Total events, bookings, page views, turnout rate, peak scanner hours graph (Recharts).
-  **Events Manager:** Create, edit, delete events, set capacity & ticket price.
-  **Guest Roster & Passes:** Attendee directory, manual check-in toggle, 1-click certificate issue & email.
-  **Certificate Template & Designer:** Drag-and-drop custom background image upload, signatory name/title customize karna, aur Live A4 preview dekhna.

 `frontend/src/pages/ClubAdminScanner.jsx`

HARDWARE QR SCANNER

✦ **Kya hai:** Gate Volunteers ke liye In-Browser Camera Scanner (HTML5-QRCode engine).

⚙️ **Kaise kaam karta hai:** Mobile/Laptop camera se student ke pass ka QR code scan karta hai aur backend par check-in validate karta hai. Duplicate ticket aane par turant Red Screen Alert deta hai.

 `frontend/src/pages/CertificateView.jsx`

A4 CERTIFICATE VIEWER

✦ **Kya hai:** Student ke certificate ka public online link (/certificate/:id).

⚙️ **Key Features:** Standard A4 Aspect Ratio (1.414:1), Landscape & Portrait view toggle, official gold authenticity seal, dynamic signatures, aur 1-click Browser Print/PDF Download (CSS @media print).

 `frontend/src/pages/SuperAdminDashboard.jsx`

MASTER GOVERNANCE

✦ **Kya hai:** College Dean / Campus Super Admin ka Master Control Panel.

⚙️ **Key Features:** Cross-Club Activity Leaderboard, Total Campus Revenue tracking, Department Distribution stats, Club suspend/activate governance, organizer password reset, aur Master E-Certificate Verification registry.



4. Top Interview Questions & Best Answers

Interview Question	How to Answer Confidently
1. How is Multi-Tenancy implemented?	<i>"We used a Shared Database, Tenant Discriminator pattern. Every document contains a <code>tenantId</code> or unique <code>sLug</code>, and tenant resolution middleware ensures complete data isolation across clubs."</i>
2. How do you prevent pass fraud at the gate?	<i>"Passes contain a unique encrypted cryptographic hash. When scanned, an atomic MongoDB update marks <code>checkedIn: true</code> with a timestamp. Any duplicate scan attempt immediately returns a 400 Bad Request error."</i>
3. How does the certificate print perfectly on A4 paper?	<i>"We render certificates as mathematical Vector SVGs with a precise 1414 x 1000 aspect ratio and standard CSS <code>@media print</code> rules, ensuring zero pixelation on physical A4 printing."</i>
4. How are large template images handled in Express?	<i>"We configured <code>express.json({ limit: '25mb' })</code> to handle high-resolution Base64 background frames seamlessly without payload limit errors."</i>

✨ Generated for Multi-Tenant Campus Club & Event Management SaaS • Production Ready